

Mini-Whiteboard Reference: Lesson 01 Square and Cube Roots

Routine: Boards down → work independently → show in 3, 2, 1 → do not erase → scan all boards. **Threshold:** about 80% correct on Q3–Q5 ⇒ move on. Below: 2–3 min reteach, then re-check.

Round 1: Calculate (after I DO) · 0:17–0:21 · slides 11–12 *SC1, SC2*

Q (purpose)	Question	Answer	Watch for	Reteach prompt
1 very easy	$\sqrt{9}$	3	4.5 (halving)	“What times itself is 9?”
2 easy	$\sqrt[3]{8}$	2	$8 \div 3$	$2 \times 2 \times 2$ cube: count 8 cubes
3 similar	$\sqrt{64}$	8	4 (cube root)	“Is there a small 3?”
4 variation	$\sqrt[3]{-125}$	-5	“no answer”, +5	$(-5) \times (-5) = 25$, $25 \times (-5) = -125$
5 misconception	T/F: $\sqrt[3]{27} = 9$	False, 3	divide by 3	“Cube 9: 729, not 27”

If below threshold: $3 \times 3 \times 3$ cube picture: “cube root asks for the *edge*”. **Re-check:** $\sqrt{36} = 6$, $\sqrt[3]{1000} = 10$.

Round 2: Between and estimate (after WE DO) · 0:28–0:31 · slides 19–20 *SC2–SC4*

Q (purpose)	Question	Answer	Watch for	Reteach prompt
1 very easy	$\sqrt{10}$ between?	3 and 4	9 and 16	“Take the root of each end”
2 easy	$\sqrt{45}$ between?	6 and 7	5 and 7	“Which squares are neighbours of 45?”
3 similar	$\sqrt[3]{20}$ between?	2 and 3	4 and 5 (squares)	“List the cubes: 1, 8, 27 ...”
4 variation	$\sqrt{30}$ to 1 d.p.	5.5	5.4; “5 to 6”	$5.4^2 = 29.16$, $5.5^2 = 30.25$
5 misconception	T/F: $\sqrt{-4} = -2$	False; not real	$(-2)^2 = -4$	$(-2)^2 = +4$; but $\sqrt[3]{-8} = -2$

If below threshold: model $\sqrt{70}$: $64 < 70 < 81$; $8 < \sqrt{70} < 9$; nearer 64; $8.3^2 = 68.89$, $8.4^2 = 70.56$; $\sqrt{70} \approx 8.4$. **Re-check:** $\sqrt{12}$ between? (3 and 4). Still wrong → start YOU DO at A1 at the front table.

Notes during the lesson